

Growing Peppers in the Dragoon Mountains

A Complete Hydroponic Pepper Growing Guide

Dragoon Mountains, Cochise County, Arizona

Home of the Native Chiltepin

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Figure 1: *The Dragoon Mountains rising above the Sulphur Springs Valley, Cochise County, Arizona.*

Chiltepin • Jalapeño • Poblano • Cayenne • Serrano • Rocoto • Habanero • Ancho • Hatch • Bell & Sweet Varieties

DWC • Dutch Bucket • NFT • Masterblend • pH • EC • Harvest

Companion guide. This guide assumes you have already read, or have access to, the two anchors:

- *Growing Peppers in the Dragoon Mountains* (Guide A) — heritage, climate, species

biology, harvest, drying, overwintering

- *Bato Bucket Hydroponic Setup* (Guide B) — the specific, field-tested pepper build

This document carries only the hydroponic deltas: system selection, reservoir management, nutrient schedule, hydro-specific pest pressure, and the hydroponic variant of overwintering. For everything else, read Guide A first — it is the root document.

Full library: A — soil peppers · B — Bato Bucket build · C — hydroponic peppers (this guide) · D — hydroponic primer · E — precision / LED · F — DIY nutrients · G — DIY pH down & up · H — grapes.

1 · Why the Dragoons Are Perfect for Hydroponic Peppers

Cool nights · monsoon humidity · diurnal swing — seen through a hydroponic lens

The Dragoon Mountains are ancestral pepper country. *Guide A* covers the Chiltepin heritage, the Apache history, and the full soil-growing climate argument. This guide assumes you know why peppers belong here. What's below is just the hydroponic re-read of the same climate.

Dragoon Factor	Hydroponic Consequence
55–70°F nights	Easy reservoir thermal control — lets you sit in the DWC sweet spot (< 72°F) with only minimal insulation outside of June.
30–40°F diurnal swing	The same swing that concentrates capsaicin in soil plants also concentrates it in hydro plants — no lighting or feeding change needed.
50–70% monsoon humidity	Lowers VPD into the hydro flowering target (1.2–1.6 kPa) for free. Most indoor growers pay for humidifiers to get here.
8–12" monsoon rainfall	Your best possible source water — collected monsoon rain sidesteps the bicarbonate problem for the second half of the season.
180+ frost-free days	Enough to run two fruiting flushes on <i>C. annuum</i> in a single outdoor hydroponic season.
15–20% higher UV	Accelerates ripening and capsaicin concentration. Outdoors beats indoors for pepper flavour — combine with LED only when you need to <i>extend</i> the season.

The soil-based Chiltepin genetics that already work in Cochise County canyons also work in a DWC net pot. Collect wild seed in fall, germinate in rockwool, and let Dragoon-adapted plants run in a passive Kratky jar on a shaded patio — zero pump, near-zero input, impressive yield.

2 · Which Peppers Do Best Hydroponically?

A system-suitability filter — not a variety encyclopaedia

For full species biology, SHU ratings, days-to-maturity, and Dragoon soil notes, see **Guide A, Chapter 2**. The table below filters those same varieties by the hydroponic system they prefer.

Variety	Best Hydro System	Why
Bell Pepper	Dutch Bucket	Bulky root mass, large canopy — needs volume. Reservoir cooling matters above 85°F.
Poblano / Ancho	Dutch Bucket ★	Thick stems, heavy fruit load. Same form factor as the Bato Bucket build in Guide B.
Anaheim / Hatch	Dutch Bucket	Similar to poblano. Long fruiting season rewards a big media volume.
Hungarian Wax	DWC or Dutch Bucket	Mid-size plant; either works.
Jalapeño	DWC ★	Classic pepper for 5-gal DWC buckets. Airstone is non-negotiable.
Serrano	DWC or NFT	Compact plant, prolific — fits well in an NFT channel if water is tight.
Chipotle (smoked jalapeño)	DWC	Grow in DWC, smoke after harvest — process is the same.
Cayenne	DWC or NFT	Thin-walled, dries perfectly outdoors after harvest.
Chiltepin (native)	NFT, Kratky, or DWC	Small, tough, perennial. Best Kratky candidate in the library.
Thai / Bird's Eye	NFT or Kratky	Small-plant / small-reservoir varieties.

Variety	Best Hydro System	Why
Habanero	Dutch Bucket ★	Needs the heat reservoir insulation can't fully solve. Bury buckets or insulate heavily.
Ghost / Bhut Jolokia	Dutch Bucket	Very long season — only Dutch Bucket holds up for 100+ days of fruiting.
Scotch Bonnet	Dutch Bucket	Same as habanero.
Rocoto (Red/Yellow)	Dutch Bucket ★	Large semi-vining plant, deeply sensitive to reservoir temp — see Chapter 3.
Manzano	Dutch Bucket	Same as Rocoto.
Aji Amarillo	Dutch Bucket	Tall plant (36–48"); needs staking <i>and</i> volume.
Lemon Drop	DWC or Dutch Bucket	Drought-tolerant in soil, forgiving in hydro.

When the table recommends Dutch Bucket — Rocoto, Habanero, Poblano, Bell, Ghost — the canonical build is the *Bato Bucket Hydroponic Setup* (Guide B). That guide carries the parts list, bench construction, RAINPOINT solar plumbing, and cost breakdown.

3 · The Rocoto Special Section — Hydroponic Edition

Reservoir temperature · in-system trellising · overwintering in-bucket

The species biology — cool-night requirement, hairy stems, black seeds, purple flowers, Andean origin — is covered in **Guide A, Chapter 3**. This section is only the hydroponic-specific layer.

Rocoto's Three Hydroponic Weak Points

Problem	Why It Matters in Hydro	Dragoon Solution
Reservoir temp > 72°F	Rocoto is cool-adapted — warm root zones cause Pythium faster than in any other pepper	Bury or insulate reservoir. Use Dutch Bucket, not DWC, above 90°F days. Place on east-facing slope so afternoon shade hits the bucket, not just the canopy.
Plant gets 5-8 ft tall and semi-vining	A 5-gallon bucket tips under that weight once fruit sets	Cage at transplant. Train main stem along a trellis or wall from day 1. Anchor bucket to bench with a strap.
Truly perennial — 15+ years	Repotting every fall wastes the root mass that makes year-2+ plants so productive	Overwinter <i>in the Dutch Bucket</i> with reduced-EC maintenance solution — see Chapter 12.

Six-Step Rocoto Hydroponic Protocol

1. Start in rockwool in January — same timing as Guide A. 10-14 weeks before net-pot transplant. Heat mat at 75-80°F.
2. Transfer to a **Dutch Bucket with perlite media** once roots emerge from the rockwool cube — not DWC unless you can guarantee reservoir temp < 72°F all summer.
3. Insulate or bury the reservoir. In the Dragoons, afternoon east-facing shade plus reservoir insulation keeps roots in the safe zone through June pre-monsoon.
4. Trellis immediately. Anchor the plant's main stem to a vertical support that can carry 10+ lb of fruit by year 2.
5. Keep EC at 1.6-2.0 at flowering. Rocoto drops flowers above EC 3.0 more readily than most varieties.
6. Overwinter in-system — see Chapter 12.

Reservoir cooling in a Dragoon summer is the single biggest hydroponic intervention available. A buried 15-gal tub stays 10-15°F cooler than the same tub sitting exposed on a patio. A wrap of 1" rigid foam is the cheapest insurance a Rocoto grower can buy.

4 • Hydroponic System Selection

DWC • Dutch Bucket • NFT • Kratky • Ebb & Flow — matched to variety and Dragoon conditions

Peppers are among the most rewarding hydroponic crops. The right system depends on the variety's size and root mass — use the Chapter 2 table as the filter. Notes below are per-system.

System Recommendations

System	Best Pepper Varieties	Container / Setup	Dragoon Notes
Dutch Bucket ★ Best for large chiles	Rocoto, Habanero, Poblano, Bell, Ghost	5-gallon buckets with perlite media. 1 plant per bucket.	Best system for the Dragoons — insulate buckets against temperature swings. Bury partially to keep root zone below 72°F. For the field-tested Dragoon build, see Guide B.
DWC (Deep Water Culture)	Jalapeño, Serrano, Cayenne, Chiltepin, Thai	5-gallon dark buckets with 3" net pots and airstone.	Simplest system. Excellent for medium peppers. Critical: keep reservoir below 72°F — insulate or bury.
NFT (Nutrient Film)	Chiltepin, Thai, Serrano, small chiles	4" channels, 12-18" plant spacing.	Best water efficiency — important in the Dragoons. Not suited for large varieties with heavy root mass.

System	Best Pepper Varieties	Container / Setup	Dragoon Notes
Kratky (Passive DWC)	Jalapeño, Serrano, Cayenne, herbs	Mason jar or 1-gallon container. No pump needed.	Excellent off-grid option. Works well outdoors in shade during monsoon. Refill every 5-7 days with fresh nutrient solution.
Ebb & Flow	Most medium varieties	Tray with timer and pump. Perlite or hydroton media.	Good for multiple smaller plants. Monitor EC carefully — solution concentrates faster in Dragoon heat.

Reservoir Temperature — Critical in the Dragoons

Root zone temperature is the #1 system management challenge in the high desert. Roots above 72°F lose dissolved oxygen, become susceptible to Pythium root rot, and stop absorbing nutrients efficiently — even when the solution is perfectly balanced.

- **■ Bury or insulate reservoirs** — underground tanks stay 10-15°F cooler. Even wrapping in foam insulation makes a significant difference.
- **■ Use dark/opaque containers** — prevents algae and reduces heat absorption.
- **■ Cover all reservoir surfaces** — evaporation in the high desert is extreme; an open 5-gallon bucket loses a gallon per day in June.
- **■ Monsoon advantage:** July-September ambient temps drop significantly — your hardest reservoir cooling period ends when peppers hit peak production.

Growing Media for Hydroponic Peppers

Media	Best System	Notes	Local Source?
Perlite (70%) + Hydroton (30%)	Dutch Bucket ★	Best drainage and aeration. Reusable. Standard commercial pepper mix.	Purchase

Media	Best System	Notes	Local Source?
Pure Perlite	Dutch Bucket, Ebb & Flow	Excellent drainage, very light. Needs more frequent irrigation cycles.	Purchase
Coarse granite grit (washed)	Dutch Bucket, DWC net pots	Free local alternative to hydroton. Rinse in dilute acid first to remove carbonates.	★ Local — crush granite
Washed local pumice	Any system	Volcanic pumice found in SE Arizona. Excellent drainage and aeration.	★ Local — Dragoon geology
Rockwool cubes (starter)	Seedling propagation	Best seedling medium for transitioning to hydroponics.	Purchase

5 • Starting Seeds & Transplanting

Rockwool starts · heat mats · transition into net pots

Peppers need a head start indoors — slow-growing and warmth-requiring. At Dragoon elevations, the last frost is April-May, so seeds started in February maximise your growing season. The key hydroponic difference: start in rockwool cubes rather than soil-based mix for a clean transition into your system.

Seed Starting Timeline by Variety

Variety Group	Start Indoors	Move to Hydro System	Reason
Rocoto / Manzano (<i>C. pubescens</i>)	January	Late April – May (4-5" tall, roots visible)	Slow germinators. Very long season (120+ days). Start earliest.
Habanero, Ghost (<i>C. chinense</i>)	Late January – Feb	Late April – May	Longest season of common varieties (85-100 days).
Jalapeño, Serrano, Cayenne (<i>C. annuum</i>)	Mid February	Late April – early May	Standard 8-10 week head start.
Poblano, Anaheim, Hatch, Bell	Mid February	Late April – early May	8-10 week head start.
Chiltepin (native)	February – March	May	Scarify seeds first (light sandpaper). Slow to establish.
Aji, Lemon Drop (<i>C. baccatum</i>)	February	Late April – May	8-10 week head start.

Germination & Propagation for Hydroponics

Factor	Target	Notes
Media	1.5" rockwool cubes ★	Soak cubes in pH 5.5 water before use. Rockwool transitions into hydro systems with no root disturbance.

Factor	Target	Notes
Substrate temperature	80–85°F (<i>C. annuum</i>); 75–80°F (Rocoto)	Use a heat mat under seed trays. Most critical germination factor.
Germination time	7–21 days (<i>C. annuum</i>); 14–28 days (<i>C. pubescens</i>)	Do not give up early on Rocoto and habanero.
Initial nutrient solution	EC 0.4–0.6 — very dilute	Seedlings need almost no nutrients. Half-strength Masterblend or plain water.
pH of water/solution	5.8–6.0	Pre-treat rockwool with pH 5.5 water (acidic). Soak 1 hour before seeding.
Light after sprouting	16 hrs LED, 200–300 PPFD	Quantum board or T5 LED 2–4" above seedlings. Keep cool to prevent stretch.

Transitioning Seedlings into the Hydroponic System

1. **Wait for visible roots** — Move to system only when roots are visible at the base of the rockwool cube — usually when seedling is 3–5" tall.
2. **Place cube in net pot** — Set rockwool cube directly in a 3" net pot. Fill around with hydroton or perlite to support. No bare-rooting needed.
3. **Drop reservoir level** — For DWC: start with water level 1" below the net pot base. Roots will grow down to find the solution within days.
4. **Start at low EC** — First week in system: EC 0.8–1.0. Increase to vegetative level (1.4–1.8) once plant shows active new growth.
5. **Protect from UV** — Dragoon UV at elevation is intense. Provide 30% shade cloth for the first 2 weeks outdoors.

6 • Water Quality & Reservoir Management

Source water • bicarbonate pre-treatment • EC • reservoir changes

SE Arizona well water is among the hardest in the country — high in dissolved calcium, magnesium, and bicarbonates that constantly push pH upward. Pre-treating your water before mixing nutrients is the single most impactful practice for stable, predictable pepper production in the Dragoons.

The 5-step bicarbonate pre-treatment is authoritative in **Guide B, Section 4**. For recipes for every acid (prickly pear vinegar, citric acid stock, phosphoric acid) and base (oak ash lye, KOH, baking soda), see **Guide G — DIY pH Down & pH Up**.

Water Source Quality

Source	Baseline EC	Suitability	Recommendation
RO filtered water	~0.0	★★★★★ Perfect	Best baseline. Mix nutrients from zero. Most consistent results.
Monsoon rainwater (collected)	< 0.05	★★★★★ Excellent	Collect July–Sept. Store in dark sealed tanks. Nearly equal to RO.
RO / rain blend 50:50	0.1–0.2	★★★★ Very Good	Stretches RO supply while maintaining low baseline.
Treated well water	0.3–0.8	★★★ Usable	Requires full bicarbonate scrub. Reduce nutrient dose to account for baseline minerals.
Untreated well water	0.3–0.8	★★ Difficult	pH will be unstable all season. Invest in RO filter for best results.

Bicarbonate Pre-Treatment — Essential for Well Water

Bicarbonates in well water buffer pH upward constantly, fighting every correction you make. Neutralise them before adding nutrients — you will see fizzing as CO₂ releases.

Wait until fizzing stops completely.

Step	Action	Target
1	Fill reservoir with source water	Note starting pH and EC
2	Add citric acid or phosphoric acid slowly while stirring	Fizzing = bicarbonates reacting and releasing CO ₂
3	Wait for fizzing to stop completely	All bicarbonates neutralised
4	Adjust pH down to 5.8-6.0	Now pH is stable — won't bounce back
5	Add Masterblend, then Epsom salt, then Calcium Nitrate	Always in this order, dissolved separately
6	Final EC and pH check	Target EC per stage table in Chapter 7

pH Targets for Hydroponic Peppers

- ■ **Target pH: 5.8 - 6.2** — tighter than soil growing. Peppers tolerate up to 6.5 but micronutrient availability drops above 6.2.
- ■ **Blossom end rot** in hydroponics is almost always a pH or EC problem, not a calcium deficiency — check these first.
- ■ Check pH daily for the first 2 weeks of a new reservoir to learn your water's drift rate.
- ■ **pH Down (local options):** prickly pear vinegar for minor adjustments; citric acid stock solution for standard corrections; phosphoric acid for pre-treating hard well water — step-by-step in Guide G.

Reservoir Management Schedule

Task	Frequency	Notes
pH check	Daily	Adjust to 5.8-6.2. Note drift direction — tells you about plant uptake.

Task	Frequency	Notes
EC check	Daily	Top off with half-strength mix if EC drops (plants drinking more water than nutrients). Top off with plain water if EC rises (evaporation concentrating the solution).
Water level top-off	Daily	Never let roots dry out. High-desert evaporation is significant — even covered reservoirs lose water.
Full reservoir change	Every 7-10 days (fruiting); every 14 days (veg)	Prevents salt buildup. Flush system with clean water between changes.
Root inspection	Weekly	Healthy = bright white. Brown/slimy = Pythium root rot — check reservoir temp first.
Meter calibration	Monthly	Calibrate pH meter with buffer solution (pH 4.0 and 7.0). Inaccurate readings are the #1 cause of unexplained crop problems.

Blossom Drop Prevention in Hydroponics

- ■ **Reservoir stability is your best tool** — consistent EC and pH prevents the nutrient stress that triggers flower drop in soil growing.
- ■ **Temperature:** Flowers drop above 90°F daytime or below 55°F nights — the Dragoon monsoon resolves both; June pre-monsoon is highest risk.
- ■ **High EC at flowering:** EC above 3.0 during bloom causes osmotic stress and blossom drop — keep EC at 1.8-2.2 through flowering.
- ■ **Thrips** damage flowers directly — yellow sticky traps near flowering plants; inspect weekly.
- ■ **Monsoon advantage:** Cooler nights in July trigger a second major flush of flowering on plants that dropped buds in June heat.

7 • Nutrient Solution Management

Masterblend programme • EC targets by stage • deficiency guide

This schedule matches the Bato Bucket Hydroponic Setup guide (Section 5) exactly. If you are building that system, use Guide B as your primary reference; this chapter is the same programme in the hydroponic-pepper-only context.

Peppers are heavy feeders with evolving nutrient needs throughout their season. The Masterblend 3-part system (4-18-38 + Epsom salt + Calcium Nitrate) delivers everything peppers need. The key is adjusting concentration — not the ratio — as the plant moves from vegetative growth to fruiting. **Peppers need more magnesium than tomatoes — double the Epsom salt dose once fruiting begins.**

Masterblend Pepper Programme — Per Gallon

Growth Stage	Masterblend (g/gal)	Epsom Salt (g/gal)	Cal-Nitrate (g/gal)	Target EC (mS/cm)	Target pH
Seedling / cutting	1.2	0.6	1.2	0.8 – 1.0	5.8 – 6.0
Early vegetative	2.4	1.2	2.4	1.4 – 1.8	5.8 – 6.0
Late vegetative	2.8	1.4	2.8	1.8 – 2.2	5.8 – 6.2
First flower buds	2.8	1.4	2.8	1.8 – 2.2	5.8 – 6.2
Peak flowering	2.4	1.2	2.4	1.6 – 2.0	5.8 – 6.0
Fruit sizing ★	3.0	2.4 ★	3.0	2.0 – 2.6	6.0 – 6.2
Peak fruiting	3.0	2.4	3.0	2.2 – 2.8	6.0 – 6.2
Late harvest	2.0	1.0	2.0	1.4 – 2.0	6.0

★ Pepper Mg rule: Peppers are the heaviest magnesium users of all common hydroponic crops. Once fruit sizing begins, increase Epsom salt to 2.4 g/gal (double the standard Masterblend ratio). Magnesium deficiency — yellowing between green veins on older leaves — is the most common pepper deficiency in hydroponics.

Nutrient Deficiency Quick Reference

Symptom	Deficiency	Hydroponic Fix	pH Cause
Yellow between green veins, older leaves first	Magnesium (Mg)	Double Epsom salt to 2.4 g/gal for 2 weeks. Foliar spray 1 tbsp/gal.	Locked out below 5.5
Yellow between green veins, new growth first	Iron (Fe) or Mn	Check pH immediately — almost always pH > 6.5. Adjust to 5.8–6.0.	Locked out above 6.5
Blossom end rot (dark spot on fruit base)	Calcium (Ca)	Check pH is 5.8–6.2. Check EC not above 3.0. Ensure Cal-Nitrate was added last and dissolved fully.	Locked out below 5.5 or above 7.0
Purple/red leaf undersides	Phosphorus (P)	Check pH not below 5.0. Warm reservoir above 65°F. Usually resolves with pH correction.	Locked out below 5.0
Pale leaves, slow growth	Nitrogen (N) — low EC	Raise overall Masterblend dose by 0.4 g/gal. Check EC is above 1.4.	N/A — EC management issue
Dark green, thick leaves, stunted	Nitrogen excess — high EC	Dilute reservoir. Do partial water change. EC should not exceed 3.0 for peppers.	N/A — EC management issue
Curled, distorted new growth	Calcium (Ca) or Boron (B)	Check mixing order — Cal-Nitrate MUST go in last. If order was correct, try a very dilute boron foliar spray.	Locked out above 7.0

EC Drift Management

- ■ **EC rising between top-offs:** Plants drinking more water than nutrients — top off with plain RO/rain water until EC returns to target.
- ■ **EC falling between top-offs:** Plants consuming nutrients heavily — top off with half-strength Masterblend mix.
- ■ **EC stable but pH rising daily:** Bicarbonate effect — pre-treat next reservoir fill with acid before adding nutrients.
- ■ **Maximum EC for peppers: 3.2 mS/cm** — above this, salt stress causes tip burn, blossom drop, and stunted fruit.

8 • Seasonal Growing Calendar

Month-by-month tasks for 4,500 – 5,500 ft hydroponic growing

Month	Key Hydroponic Tasks	System Status
January	Start Rocoto and habanero in rockwool under LEDs. Set up heat mats. Mix dilute seedling solution (EC 0.6). Clean and test all meters.	Rocoto in rockwool; LED seedling stage
February	Start all <i>C. annuum</i> varieties in rockwool. Set up and test reservoirs. Order Masterblend, Epsom salt, and Calcium Nitrate. Order RO filter if needed.	Pepper seedlings under grow lights
March	Pot up seedlings as roots emerge from rockwool. Begin vegetative nutrient solution (EC 1.4–1.8). Inspect system for leaks. Set up monsoon water collection.	Seedlings 4–6" tall in net pots
April	Transition outdoor systems. Insulate/bury reservoirs before heat arrives. Install shade cloth. Watch for late frost — bring indoors if below 40°F.	Moving to outdoor systems
May	All plants in outdoor systems. EC at vegetative level (1.4–1.8). Daily pH and EC checks. Stake large varieties immediately.	Rapid vegetative growth

Month	Key Hydroponic Tasks	System Status
June	First flower buds — reduce EC slightly to 1.8-2.2. Reservoir management critical in pre-monsoon heat. Check reservoir temp daily — must stay below 72°F.	Flowering; blossom drop risk
July	MONSOON BEGINS. High humidity — watch for powdery mildew and aphid surges. Increase Epsom salt to 2.4 g/gal (fruit sizing). Rocoto flowering heavily — perfect conditions.	Fruit sizing; monsoon rain
August	Peak production — <i>C. annuum</i> harvest begins. Change reservoirs every 7 days during peak fruiting. EC at 2.2-2.8 for fruiting stage.	Harvest jalapeños, serranos, cayenne
September	★ BEST MONTH. Cool nights return — plant vigour surges. Harvest all fully ripe varieties. Rocoto at peak. Change reservoir weekly. Keep EC at fruiting level.	Peak harvest all varieties
October	Harvest continues. Frost watch begins. Move container systems indoors before first frost. Rocoto — pot up and bring inside. Begin transitioning to LED indoor winter grow if desired.	Final harvest; indoor transition

Month	Key Hydroponic Tasks	System Status
November	Overwinter selected plants indoors under LEDs. Reduce EC to 1.2-1.4 (semi-dormant maintenance). Dry and process harvest. Clean outdoor system for storage.	Indoor overwintering; preservation
December	Overwintered plants under LED at 16 hrs. EC 1.0-1.4 — minimal feeding. Plan next season. Order seeds. Review reservoir notes from past season.	Semi-dormant indoor growing

9 • Environmental Control & Canopy

VPD • temperature • LED lighting • staking • increasing capsaicin

Hydroponic peppers respond dramatically to environmental optimisation. The Dragoon Mountains give you naturally good VPD in fall, cool nights for fruit set, and high-altitude UV for flavour development — advantages no indoor-only grower can replicate. Combine these with good canopy management for exceptional yields.

VPD Targets for Hydroponic Peppers

Stage	Temp	Humidity	VPD (kPa)	Dragoon Season
Seedling	72-77°F	70-80%	0.4-0.8	Indoor Jan-Mar
Early vegetative	72-82°F	60-70%	0.8-1.1	Indoor/outdoor Apr-May
Late veg / pre-flower	75-85°F	55-65%	1.0-1.4	May-June
Flowering	70-82°F	50-60%	1.2-1.6	June-July (monsoon helps)
Peak fruiting	75-85°F	50-60%	1.2-1.6	July-Sept (ideal naturally)
Late harvest	65-78°F	45-55%	1.4-1.8	Sept-Oct (Dragoon fall peak)

LED Lighting for Indoor & Greenhouse Pepper Growing

Scenario	PPFD Target	DLI	Photoperiod	Notes
Indoor seedlings	150-250 $\mu\text{mol}/\text{m}^2/\text{s}$	8-14	16-18 hrs	T5 or small quantum board 4" above canopy.
Indoor vegetative	400-600 $\mu\text{mol}/\text{m}^2/\text{s}$	20-30	16-18 hrs	Spider Farmer SF-2000 or equivalent. 18-24" above canopy.
Indoor fruiting	600-800 $\mu\text{mol}/\text{m}^2/\text{s}$	25-35	14-16 hrs	High-output quantum board. Drop to 16 hrs to simulate seasonal change.

Scenario	PPFD Target	DLI	Photoperiod	Notes
Greenhouse supplement	200-400 $\mu\text{mol}/\text{m}^2/\text{s}$ (supplemental)	Adds 8-15 to natural DLI	2-4 hrs extension only	LED bar lights at dawn/dusk to extend photoperiod. Very cost effective.
Rocoto (summer)	200-400 $\mu\text{mol}/\text{m}^2/\text{s}$ (filtered)	12-18	14-16 hrs	Rocoto needs shade in summer. 30-40% shade cloth + LED in morning/evening gives ideal levels.

Canopy Management for Hydroponic Peppers

- **■ Stake at transplant to net pot** — use a bamboo cane in the bucket/net pot alongside the plant from day one.
- **■ Remove first flower buds for 2-3 weeks** — redirects energy to root development in the hydroponic media, significantly improving total season yield.
- **■ Pinch growing tip at 6-8" height** — forces branching and doubles fruiting sites.
- **■ Remove leaves below the first fork** — improves airflow around reservoir opening, reducing algae risk.
- **■ Rocoto training:** train the main stem along a trellis — these become large semi-vining plants in hydroponic systems and can produce hundreds of fruits in year 2+.

How to Maximise Capsaicin in Hydroponics

- **■ Controlled water stress, final 2 weeks:** reduce reservoir topping-off frequency slightly. Mild stress — roots still wet — significantly increases capsaicin without damaging the plant.
- **■ Reduce nitrogen after fruit set:** drop Masterblend from 3.0 to 2.0 g/gal in the last 3 weeks while maintaining Epsom salt and Cal-Nitrate at full rate.
- **■ High PPFD accelerates ripening and capsaicin concentration:** increase to 700-800 $\mu\text{mol}/\text{m}^2/\text{s}$ in final 2-3 weeks.
- **■ Let fruit ripen fully:** red jalapeños have 2-3× the capsaicin of green ones. The extra 3-4 weeks is worth it.

- ■ **Grow native Chiltepin:** the most intense local variety, thrives in DWC, and produces prolifically with minimal inputs.

10 • Pest & Disease Management

Hydro-specific pressure • reservoir-borne problems • organic controls

The soil pests — cutworms at transplant time, javelina uprooting a bed, grasshoppers in drought years — live in **Guide A, Chapter 10**. They apply only to outdoor beds. Hydroponic systems have their own pressure set, dominated by root-zone pathogens and canopy pests.

Pest / Problem	Season	Signs	Hydroponic Control
Pythium root rot ★ #1 hydro threat	Peak summer — reservoirs > 72°F	Brown/slimy roots; wilting despite full reservoir; sour smell from solution	Drop reservoir temperature below 72°F — bury, insulate, or shade. Flush system. Dose 3 ml/gal 3% hydrogen peroxide. Replace media if severe.
Aphids ★ Most common	Spring & Fall	Dense clusters on new growth; sticky honeydew; ants farming them	Strong water blast. Insecticidal soap spray. Neem oil. Plant basil nearby. Ladybugs (buy or attract naturally).
Spider mites	June (hot dry)	Tiny webbing under leaves; bronzed, stippled foliage	Increase humidity around plants (mist in morning). Neem oil or insecticidal soap. Avoid broad-spectrum pesticides.
Thrips	Spring–Summer	Silvery scars on leaves and fruit; distorted new growth; dropped flowers	Yellow sticky traps. Spinosad spray. Remove affected leaves. Blue sticky traps specifically target thrips.

Pest / Problem	Season	Signs	Hydroponic Control
Pepper weevil	Summer	Small holes in green fruit; fruit drops prematurely; internal larvae	Remove dropped fruit immediately. Neem oil spray on developing fruit. Pyrethrin as last resort.
Algae in reservoir	Year-round, worse in summer	Green film on reservoir walls and net pots; slimy net pots	Cover reservoir completely. Use opaque black/blue tubs. Clean between reservoir changes.
Fungus gnats	Cool season in transitioning rockwool	Small black flies around net pots; larvae in rockwool	Top-dress with sand or perlite. BTi (Gnatrol) drench on rockwool. Yellow sticky traps.
Blossom end rot	Peak fruiting	Dark sunken spot on fruit bottom	See Chapter 7 — pH 5.8-6.2 + EC < 3.0 + Cal-Nitrate last. Nearly always a dosing/pH problem, not a Ca deficit.
Sun-scald	June-August	White papery patches on fruit facing the sun	30% shade cloth in peak heat. Leave enough foliage to shade fruit. Orient rows north-south so plants shade each other.

Pest / Problem	Season	Signs	Hydroponic Control
Powdery mildew	Monsoon season	White powder on leaves — mainly in dense canopy	Improve airflow around plants. Dilute milk spray (1 part milk : 9 parts water). Potassium bicarbonate spray. Increase plant spacing.

If Your System Is Outdoors

Hydroponic systems set up on an open patio or field get the same outdoor-pest pressure that soil plants do. When that's the case, the following soil threats apply — consult **Guide A, Chapter 10** for treatment:

- Cutworms (at transplant time) — a cardboard collar around each plant stem is enough.
- Grasshoppers (June–Sept) — especially in drought years.
- Javelina (year-round) — electric fence the bed perimeter.
- Bacterial wilt — remove affected plant and discard media; rotate the system location if possible.

11 • Harvest, Drying & Processing

Harvest timing, green-versus-red tradeoffs, drying methods, ristra construction, and mesquite-smoked chipotle are identical for soil-grown and hydroponic peppers. The protocol is authoritative in **Guide A, Chapter 11** — see that chapter for the full harvest table, the five drying methods, and the Dragoon mesquite smoking notes.

The only hydroponic-specific note: at the end of a hydroponic plant's life, run the end-of-crop H_2O_2 + vinegar flush described in **Guide B, Section 7** before the next crop. Everything else is the same.

12 • Overwintering Peppers — Hydroponic Protocol

Keep them in the bucket — not the pot

Which varieties are worth overwintering, the hard cut-back, pest inspection before bringing indoors, hardening off in May — all covered in **Guide A, Chapter 12**. That chapter applies equally to soil and hydro plants. What follows is only the hydroponic-specific variant: *keep the plant in its Dutch bucket or DWC container all winter rather than repotting*.

Hydroponic Overwintering Protocol

1. **Cut back hard in early October** — Same as soil: prune to 6-8" stubs, 3-4 main branches. Reduces transpiration and makes the plant easy to move.
2. **Inspect and treat for pests** — Check carefully for mites, aphids, and scale before bringing inside. Insecticidal soap, let dry.
3. **Keep plant in its hydroponic container** — Leave Rocoto and habanero in their Dutch buckets with perlite media. No repotting. Root mass stays intact — this is the whole point of overwintering in-system.
4. **Switch to maintenance solution** — EC 1.0-1.2 with the standard Masterblend ratio. Change reservoir every 3-4 weeks over winter. Plants are semi-dormant; uptake is minimal.
5. **Reduce light and water** — 16 hrs LED at reduced intensity (PPFD 200-300). Top off reservoir weekly rather than daily.
6. **Resume feeding in February** — Bring EC back to vegetative level (1.4-1.8). Plants will push new growth rapidly — year-2 Rocoto especially.
7. **Harden off and return outdoors in May** — Same 1-2 week acclimatisation as a seedling. Year-2 hydroponic Rocoto in the original Dutch bucket will explode into production within weeks.

Hydroponics is how you put Andean conditions inside a Cochise County reservoir. The Dragoon air gives you cool nights, monsoon humidity, and high-altitude UV for free — the DWC bucket, Dutch bucket, and NFT channel just stabilise everything beneath the canopy. With the water pre-treated, the root zone kept below 72°F, and the Masterblend schedule tracked to the pepper Mg rule, a hydroponic grower on a patio in Cochise County can grow Rocoto at the same latitude the Apache once harvested Chiltepin — and the peppers will notice.